AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 8: LU Factorization; Gaussian Elimination with Pivoting

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Outline

1. **Gaussian Elimination (MC §3.2, NLA §20)**
2. Gaussian Elimination with Pivoting (NLA §21)

Review of Gaussian Elimination

- Given linear system $Ax = b$, where $A \in \mathbb{R}^{n \times n}$ is nonsingular, Gaussian elimination transforms the linear system to

$$Ux = y$$

  where U is upper triangular, and then solves $Ux = y$ by back substitution.
- Gaussian elimination performs three types of elementary row operations:
  1. Adding a multiple of one equation to another.
  2. Interchanging two equations.
  3. Multiplying an equation by a nonzero constant.
- Gaussian elimination can be represented as transformation of augmented matrix

$$[A \mid b] \rightarrow [U \mid y]$$

  where the operations are on rows of this matrix.
- A is nonsingular if and only if U is nonsingular.

Example of Gaussian Elimination

Consider the linear system

$$\begin{aligned} 3x_1 + 5x_2 &= 9 \\ 6x_1 + 7x_2 &= 4 \end{aligned}$$

- By multiplying the first row by 2 (i.e., $2 = 6/3$) and subtracting it from the second row, we obtain

$$\begin{aligned} 3x_1 + 5x_2 &= 9 \\ -3x_2 &= -14 \end{aligned}$$

- The upper triangular system can then be solved using back substitution.

LU Factorization

- For $Ax = b$, suppose we can find $A = LU$, where L is lower triangular and U is upper triangular. Then solve the system in three steps:
  1. Perform LU factorization $A = LU$.
  2. Solve $Ly = b$ using forward substitution.
  3. Solve $Ux = y$ using back substitution.
- In the previous example,

$$\underbrace{\begin{bmatrix} 3 & 5 \\ 6 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 3 & 5 \\ 0 & -3 \end{bmatrix}}_{U}$$

- LU factorization and Gaussian elimination are often used interchangeably.
- Note: $A = LU$ may not always exist even if A is nonsingular.
  - Why? Division by zero may occur.
  - Remedy: Pivoting may be needed.
- We start with the simplest case without pivoting.

Gaussian Elimination and LU Factorization

- Gaussian elimination without row interchanges can be viewed as "triangularization" of *nonsingular* $A \in \mathbb{R}^{n\times n}$:

$$\underbrace{L_{n-1}\cdots L_2 L_1}_{L^{-1}} A = U$$

- Then $A = LU$. It is also called LU factorization (without pivoting).
- For the augmented matrix, $L^{-1}[A \mid b] = [U \mid y]$, so $y = L^{-1}b$.
- Example of *LU* factorization of 4×4 matrix A:

$$\xrightarrow{L_1} \underbrace{\begin{bmatrix} \times & \times & \times & \times \\ 0 & \mathbf{x} & \mathbf{x} & \mathbf{x} \\ 0 & \mathbf{x} & \mathbf{x} & \mathbf{x} \\ 0 & \mathbf{x} & \mathbf{x} & \mathbf{x} \end{bmatrix}}_{L_1 A} \xrightarrow{L_2} \underbrace{\begin{bmatrix} \times & \times & \times & \times \\ & \times & \times & \times \\ & 0 & \mathbf{x} & \mathbf{x} \\ & 0 & \mathbf{x} & \mathbf{x} \end{bmatrix}}_{L_2 L_1 A} \xrightarrow{L_3} \underbrace{\begin{bmatrix} \times & \times & \times & \times \\ & \times & \times & \times \\ & & \times & \times \\ & & 0 & \mathbf{x} \end{bmatrix}}_{L_3 L_2 L_1 A}$$

Example of LU Factorization

- Consider

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 3 & 3 & 1 \\ 8 & 7 & 9 & 5 \\ 6 & 7 & 9 & 8 \end{bmatrix}$$

- Then $L_1 = \begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ -4 & & 1 & \\ -3 & & & 1 \end{bmatrix}$ and $L_1 A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ & 1 & 1 & 1 \\ & 3 & 5 & 5 \\ & 4 & 6 & 8 \end{bmatrix}$

Example of LU Factorization Cont'd

- $L_1 A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ & 1 & 1 & 1 \\ & 3 & 5 & 5 \\ & 4 & 6 & 8 \end{bmatrix}$
- $L_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & -3 & 1 & 0 \\ & -4 & 0 & 1 \end{bmatrix}$ and $L_2 L_1 A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ & 1 & 1 & 1 \\ & & 2 & 2 \\ & & 2 & 4 \end{bmatrix}$
- $L_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & -1 & 1 \end{bmatrix}$ and $L_3 L_2 L_1 A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ & 1 & 1 & 1 \\ & & 2 & 2 \\ & & 0 & 2 \end{bmatrix}$

Example of LU Factorization Cont'd

- $L_1 = \begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ -4 & & 1 & \\ -3 & & & 1 \end{bmatrix}$, $L_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & -3 & 1 & 0 \\ & -4 & 0 & 1 \end{bmatrix}$, $L_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & -1 & 1 \end{bmatrix}$ and

$$L_3 L_2 L_1 A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ & 1 & 1 & 1 \\ & & 2 & 2 \\ & & 0 & 2 \end{bmatrix}$$

- Then,

$$\underbrace{\begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 3 & 3 & 1 \\ 8 & 7 & 9 & 5 \\ 6 & 7 & 9 & 8 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & & & \\ 2 & 1 & & \\ 4 & 3 & 1 & \\ 3 & 4 & 1 & 1 \end{bmatrix}}_{L=L_1^{-1}L_2^{-1}L_3^{-1}} \underbrace{\begin{bmatrix} 2 & 1 & 1 & 0 \\ & 1 & 1 & 1 \\ & & 2 & 2 \\ & & 0 & 2 \end{bmatrix}}_{U}.$$

What is Matrix L_1?

- In Gaussian elimination, at the first step A is transformed to $A^{(1)}$ by

$$\ell_{i1} = a_{i1}/a_{11}, \qquad i = 2, \dots, n$$
$$a_{ij}^{(1)} = a_{ij} - \ell_{i1} a_{1j}, \qquad i = 2, \dots, n, \quad j = 2, \dots, n$$

- In matrix form, we have $A^{(1)} = L_1 A$, where

$$L_1 = \begin{bmatrix} 1 & & & & \\ -\ell_{2,1} & 1 & & & \\ -\ell_{3,1} & & 1 & & \\ \vdots & & & \ddots & \\ -\ell_{n,1} & & & & 1 \end{bmatrix}$$

Structure of the Matrix L_k

- At step k, eliminate entries below $a_{kk}^{(k-1)}$.
- Let $a_{:,k}^{(k-1)}$ be the kth column of $L_{k-1} \cdots L_1 A$, then

$$a_{:,k}^{(k-1)} = [a_{1k}^{(k-1)}, a_{2k}^{(k-1)}, \cdots, a_{kk}^{(k-1)}, a_{k,k+1}^{(k-1)}, \cdots a_{n,k}^{(k-1)}]^\top$$

$$L_k a_{:,k}^{(k-1)} = [a_{1k}^{(k-1)}, a_{2k}^{(k-1)}, \cdots, a_{kk}^{(k-1)}, 0, \cdots 0]^\top$$

- The *multipliers* $\ell_{jk} = a_{jk}^{(k-1)} / a_{kk}^{(k-1)}$ appear in L_k:

$$L_k = \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & -\ell_{k+1,k} & 1 & & \\ & & \vdots & & \ddots & \\ & & -\ell_{n,k} & & & 1 \end{bmatrix}$$

Forming The L Matrix

- Let $\ell_k = [\underbrace{0, \cdots, 0}_{k}, \ell_{k+1,k}, \cdots, \ell_{n,k}]^\mathsf{T}$ and $e_k = [\underbrace{0, \cdots, 0}_{k-1}, 1, \cdots, 0]^\mathsf{T}$, then $L_k = I - \ell_k e_k^\mathsf{T}$.
- Luckily, L matrix contains the multipliers $\ell_{jk} = x_{jk}/x_{kk}$

$$L = L_1^{-1} L_2^{-1} \cdots L_{n-1}^{-1} = \begin{bmatrix} 1 & & & & \\ \ell_{21} & 1 & & & \\ \ell_{31} & \ell_{32} & 1 & & \\ \vdots & \vdots & \ddots & \ddots & \\ \ell_{n1} & \ell_{n2} & \cdots & \ell_{n,n-1} & 1 \end{bmatrix}$$

and is said to be a *unit lower triangular matrix*.

Proof of Structure of L Matrix

- First, $L_k^{-1} = I + \ell_k e_k^\top$, because $e_k^\top \ell_k = 0$ and

$$(I - \ell_k e_k^\top)(I + \ell_k e_k^\top) = I - \ell_k e_k^\top \ell_k e_k^\top = I$$

- Second, $L_1^{-1} L_2^{-1} \cdots L_{k+1}^{-1} = I + \sum_{j=1}^{k+1} \ell_j e_j^\top$, since (prove by induction)

$$(I + \sum_{j=1}^{k} \ell_j e_j^\top)(I + \ell_{k+1} e_{k+1}^\top) = I + \sum_{j=1}^{k+1} \ell_j e_j^\top + \sum_{j=1}^{k} \ell_j (e_j^\top \ell_{k+1}) e_{k+1}^\top$$

  where $e_j^\top \ell_{k+1} = 0$ for $j < k+1$.

- In other words, L is "union" of nonzero entries in $L_1^{-1}, L_2^{-1}, \cdots, L_{n-1}^{-1}$.

LU Factorization without Pivoting

- Factorize $A \in \mathbb{R}^{n\times n}$ into $A = LU$.

Gaussian elimination without pivoting

$U \leftarrow A, L \leftarrow I$
for $k = 1 : n-1$
  for $j = k+1 : n$
    $\ell_{jk} \leftarrow u_{jk}/u_{kk}$
    $u_{j,k:n} \leftarrow u_{j,k:n} - \ell_{jk} u_{k,k:n}$

- Flop count $\sim \sum_{k=1}^{n} 2(n-k)(n-k) \sim 2\sum_{k=1}^{n} k^2 \sim 2n^3/3$.
- In practice, L overwrites lower triangular part of A and U overwrites upper triangular part of A.
- Question: What happens if u_{kk} is 0?
- Answer: The algorithm will break due to division by zero!

Theorem of LU Factorization

Theorem

Let A be an $n \times n$ matrix whose leading principal submatrices are all nonsingular. Then A can be factorized in exactly one way into a product $A = LU$, such that L is unit lower triangular and U is upper triangular.

- However, the requirement of nonsingular leading principal submatrices is too strong!
- More importantly, what happens if u_{kk} is ***nearly*** 0 (or a submatrix is ***nearly*** singular)?

Outline

1. Gaussian Elimination (MC §3.2, NLA §20)
2. **Gaussian Elimination with Pivoting (NLA §21)**

Gaussian Elimination with Partial Pivoting

- At step k, we divide by u_{kk} (i.e., $a_{kk}^{(k-1)}$), which would break if u_{kk} is 0 (or close to 0), which can happen even if A is nonsingular.
- Other nonzero entry in kth column below diagonal can be *pivot*

$$\begin{bmatrix} \times & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & a_{ik}^{(k-1)} & \mathbf{x} & \mathbf{x} & \mathbf{x} \\ & \times & \times & \times & \times \end{bmatrix} \rightarrow \begin{bmatrix} \times & \times & \times & \times & \times \\ & 0 & \mathbf{x} & \mathbf{x} & \mathbf{x} \\ & 0 & \mathbf{x} & \mathbf{x} & \mathbf{x} \\ & a_{ik}^{(k-1)} & \times & \times & \times \\ & 0 & \mathbf{x} & \mathbf{x} & \mathbf{x} \end{bmatrix}$$

and we permute (interchange) row i with row k.

Partial Pivoting

- We choose nonzero entry with largest magnitude as pivot.
- kth step of Gaussian elimination of partial pivoting

$$\begin{bmatrix} \times & \times & \times & \times \\ & \times & \times & \times \\ & x_{ik} & \mathbf{x} & \mathbf{x} \\ & \times & \times & \times \end{bmatrix} \xrightarrow{P_k} \begin{bmatrix} \times & \times & \times & \times \\ & x_{kk} & \mathbf{x} & \mathbf{x} \\ & \mathbf{x} & \mathbf{x} & \mathbf{x} \\ & \times & \times & \times \end{bmatrix} \xrightarrow{L_k} \begin{bmatrix} \times & \times & \times & \times \\ & x_{kk} & \times & \times \\ & \mathbf{0} & \mathbf{x} & \mathbf{x} \\ & \mathbf{0} & \mathbf{x} & \mathbf{x} \end{bmatrix}$$

Pivot selection Row interchange Elimination

  and we interchange row i with row k.
- P_k is a permutation matrix, obtained by interchanging two rows of I.
- The product of permutation matrices is also a permutation matrix.
- For any permutation matrix P, all its entries are zeros and ones, and $PP^{\mathsf{T}} = P^{\mathsf{T}}P = I$.

Example of LU with Partial Pivoting

- Consider $A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 3 & 3 & 1 \\ 8 & 7 & 9 & 5 \\ 6 & 7 & 9 & 8 \end{bmatrix}$
- $\begin{bmatrix} & & 1 & \\ & 1 & & \\ 1 & & & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 3 & 3 & 1 \\ 8 & 7 & 9 & 5 \\ 6 & 7 & 9 & 8 \end{bmatrix} = \begin{bmatrix} 8 & 7 & 9 & 5 \\ 4 & 3 & 3 & 1 \\ 2 & 1 & 1 & 0 \\ 6 & 7 & 9 & 8 \end{bmatrix}$
- $L_1 = \begin{array}{c} P_1 \\ \begin{bmatrix} 1 & & & \\ -\frac{1}{2} & 1 & & \\ -\frac{1}{4} & & 1 & \\ -\frac{3}{4} & & & 1 \end{bmatrix} \end{array}$ and $L_1 P_1 A = \begin{bmatrix} 8 & 7 & 9 & 5 \\ & -\frac{1}{2} & -\frac{3}{2} & -\frac{3}{2} \\ & -\frac{3}{4} & -\frac{5}{4} & -\frac{5}{4} \\ & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \end{bmatrix}$

Example of LU with Partial Pivoting Cont'd

- $$\underbrace{\begin{bmatrix} 1 & & & \\ & & & 1 \\ & & 1 & \\ & 1 & & \end{bmatrix}}_{P_2} \underbrace{\begin{bmatrix} 8 & 7 & 9 & 5 \\ & -\frac{1}{2} & -\frac{3}{2} & -\frac{3}{2} \\ & -\frac{3}{4} & -\frac{5}{4} & -\frac{5}{4} \\ & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \end{bmatrix}}_{L_1P_1A} = \underbrace{\begin{bmatrix} 8 & 7 & 9 & 5 \\ & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \\ & -\frac{3}{4} & -\frac{5}{4} & -\frac{5}{4} \\ & -\frac{1}{2} & -\frac{3}{2} & -\frac{3}{2} \end{bmatrix}}_{P_2L_1P_1A}$$

- $$L_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & \frac{3}{7} & 1 & 0 \\ & \frac{2}{7} & 0 & 1 \end{bmatrix} \text{ and } L_2P_2L_1P_1A = \begin{bmatrix} 8 & 7 & 9 & 5 \\ & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \\ & & -\frac{2}{7} & \frac{4}{7} \\ & & -\frac{6}{7} & -\frac{2}{7} \end{bmatrix}$$

Example of LU with Partial Pivoting Cont'd

- $$\underbrace{\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & & 1 \\ & & 1 & \end{bmatrix}}_{P_3} \underbrace{\begin{bmatrix} 8 & 7 & 9 & 5 \\ & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \\ & & -\frac{2}{7} & \frac{4}{7} \\ & & -\frac{6}{7} & -\frac{2}{7} \end{bmatrix}}_{L_2P_2L_1P_1A} = \underbrace{\begin{bmatrix} 8 & 7 & 9 & 5 \\ & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \\ & & -\frac{6}{7} & -\frac{2}{7} \\ & & -\frac{2}{7} & \frac{4}{7} \end{bmatrix}}_{P_3L_2P_2L_1P_1A}$$
- $L_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & -\frac{1}{3} & 1 \end{bmatrix}$ and $L_3P_3L_2P_2L_1P_1A = \begin{bmatrix} 8 & 7 & 9 & 5 \\ & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \\ & & -\frac{6}{7} & -\frac{2}{7} \\ & & & \frac{2}{3} \end{bmatrix}$

Example of LU with Partial Pivoting Cont'd

- Then,

$$\underbrace{\begin{bmatrix} & & 1 & \\ & & & 1 \\ & 1 & & \\ 1 & & & \end{bmatrix}}_{P}\underbrace{\begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 3 & 3 & 1 \\ 8 & 7 & 9 & 5 \\ 6 & 7 & 9 & 8 \end{bmatrix}}_{A}$$

$$= \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{3}{4} & 1 & 0 & 0 \\ \frac{1}{2} & -\frac{2}{7} & 1 & 0 \\ \frac{1}{4} & -\frac{3}{7} & \frac{1}{3} & 1 \end{bmatrix}}_{L}\underbrace{\begin{bmatrix} 8 & 7 & 9 & 5 \\ & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \\ & & -\frac{6}{7} & -\frac{2}{7} \\ & & & \frac{2}{3} \end{bmatrix}}_{U}.$$

Matrix Notation of Partial Pivoting

- In terms of matrices, it becomes $\underbrace{L_{n-1}P_{n-1}\cdots L_2P_2L_1P_1}_{L^{-1}P}A = U$.
- $PA = LU$ or $A = P^\top LU$. This is LU factorization with partial pivoting.
- $P = P_{n-1}\cdots P_2P_1$ and $L = (L'_{n-1}\cdots L'_2L'_1)^{-1}$, where $L'_k = P_{n-1}\cdots P_{k+1}L_kP_{k+1}^{-1}\cdots P_{n-1}^{-1}$.
- It can be verified that $L_{n-1}P_{n-1}\cdots L_2P_2L_1P_1 = (L'_{n-1}\cdots L'_2L'_1)(P_{n-1}\cdots P_2P_1)$.
- $L'_k = I - P_{n-1}\cdots P_{k+1}\ell_ke_k^\top$, and $\left(L'_k\right)^{-1} \equiv I + P_{n-1}\cdots P_{k+1}\ell_ke_k^\top$.
- In other words, L'_k is obtained by continued permutation of $\ell_ke_k^\top$ along with $A^{(k+)}$, and lower triangular part of L is "union" of permuted $\ell_ke_k^\top$.

Algorithm of Gaussian Elimination with Partial Pivoting

- Factorize $A \in \mathbb{R}^{n\times n}$ into $A = P^{\top}LU$.

Gaussian elimination with partial pivoting

$U \leftarrow A, L \leftarrow I, P \leftarrow I$
for $k = 1 : n-1$
  $i \leftarrow \arg\max_{i\geq k} |u_{ik}|$
  $u_{k,k:n} \leftrightarrow u_{i,k:n}$
  $\ell_{k,1:k-1} \leftrightarrow \ell_{i,1:k-1}$
  $p_{k,:} \leftrightarrow p_{i,:}$
  for $j = k+1 : n$
    $\ell_{jk} \leftarrow u_{jk}/u_{kk}$
    $u_{j,k:n} \leftarrow u_{j,k:n} - \ell_{jk}u_{k,k:n}$

- Flop count $\sim \sum_{k=1}^{n} 2(n-k)(n-k) \sim 2\sum_{k=1}^{n} k^2 \sim 2n^3/3$, same as without pivoting.
- Question: Can u_{kk} be 0?

An Alternative Implementation

- In practice, L and U overwrite A, and P is represented by a vector.

Gaussian elimination with partial pivoting (alternative)

$p \leftarrow [1, 2, \cdots, n]$
for $k = 1 : n-1$
  $i \leftarrow \arg\max_{i \geq k} |a_{ik}|$
  $a_{k,1:n} \leftrightarrow a_{i,1:n}$
  $p_k \leftrightarrow p_i$
  $a_{k+1:n,k} \leftarrow a_{k+1:n,k} / a_{k,k}$
  $A_{k+1:n,k+1:n} \leftarrow A_{k+1:n,k+1:n} - a_{k+1:n,k} * a_{k,k+1:n}$

- Using LU factorization with partial pivoting to solve $Ax = b$:
  1. $A = P^{\mathsf{T}} LU$; (LU factorization with partial pivoting).
  2. $Ly = Pb$; (Forward substitution, where $(Pb)_i = b(p_i)$).
  3. $Ux = y$; (Back substitution).
- If the augmented matrix is used, then the first two steps are merged.

Complete Pivoting

- More generally, the pivot can be chosen from entries (i, j), $i \geq k$, $j \geq k$

$$\begin{bmatrix} \times & \times & \times & \times \\ & \times & \times & \times \\ & \mathbf{x} & x_{ij} & \mathbf{x} \\ & \times & \times & \times \end{bmatrix} \rightarrow \begin{bmatrix} \times & \times & \times & \times \\ & \mathbf{x} & \mathbf{0} & \mathbf{x} \\ & \times & x_{ij} & \times \\ & \mathbf{x} & \mathbf{0} & \mathbf{x} \end{bmatrix}$$

  and we then permute row i with row k, column j with column k.

- In matrix operations, complete pivoting can be expressed as

$$\underbrace{L_{n-1}P_{n-1} \cdots L_2P_2L_1P_1}_{L^{-1}P} A \underbrace{Q_1Q_2 \cdots Q_{n-1}}_{Q} = U$$

- Therefore, $PAQ = LU$ where $P = P_{n-1} \cdots P_2P_1$ and $L = (L'_{n-1} \cdots L'_2L'_1)^{-1}$.

Complete Pivoting

- By choosing largest absolute value among these entries, complete pivoting gives better stability theoretically (to be discussed later).
- Complete pivoting is typically not used in practice, because it increases cost in searching pivot and complexity of implementation.
  - Complete pivoting incurs $\mathcal{O}\left(n^3\right)$ comparisons, whereas partial pivoting incurs $\mathcal{O}\left(n^2\right)$.
- Complete pivoting can be useful in solving underdetermined systems.
  - System is underdetermined if A is singular, and it has an infinite number of solutions.
  - One can compute
    $$PAQ^{\mathsf{T}} = L\left[U_1 \mid U_2\right],$$
    where U_1 is nonsingular and upper triangular, and U_2 has zeros along diagonal.
  - It allows finding a solution, but not necessarily an optimal solution.

Alternative Linear Solvers

Gauss–Jordan elimination: At the kth step, use pivot to eliminate nonzeros in column k both above and below diagonal. It is more expensive than Gaussian elimination. How much more?

Cramer's rule: $x_i = \det(A^{(i)})/\det(A)$, where $A^{(i)}$ is obtained from A by replacing ith column by b. This is overly expensive and unstable, so it is never used except for $n = 2$ or 3.

Strassen's method: (and similar recursive algorithms) for matrix-matrix multiplication can be modified to solve linear systems in $\mathcal{O}(n^s)$ flops for $s < 3$, but are not advantageous in practice.

These are direct methods. Other important classes of methods include

- *iterative methods* (later) and
- *multigrid methods*.